Lecture 2: Random variables – Part 1

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Random Variables

Definition

*A **random variable** (RV) is a real-valued function defined on a sample space.*

There are

- ***discrete*** random variables – accept values from ***countable finite or infinite*** set.
- ***continuous*** random variables – accept values from ***non-countable*** set (finite, semi-infinite, or infinite interval).

Discrete random variables

Countable finite

X	x_1	x_2	$\dots$	x_n
p	p_1	p_2	$\dots$	p_n

$$\sum_{i=1}^{n} p_i = 1.$$

Countable infinite

X	x_1	x_2	$\dots$	x_n	$\dots$
p	p_1	p_2	$\dots$	p_n	$\dots$

$$\sum_{i=1}^{+\infty} p_i = 1.$$

The tables above are examples of a ***distribution of a random variable*** X. Also, they define a ***probability mass function***

$$p_i = \mathrm{P}\left(X = x_i\right),\ i = 1, 2, \dots.$$

Numerical characteristics of discrete random variables

Expected value

An ***expected value*** (μ) of a random variable represents the average of a large number of independent realizations of the random variable.

The ***expected value*** of a discrete random variable is the probability-weighted average of all its possible values:

$$\mu = \mathrm{E}\left[X\right] = \sum_i x_i p_i.$$

The expected value is also known as the ***expectation***, ***mathematical expectation***, ***mean***, or ***first moment***.

For example, the expected value of rolling a six-sided die is 3.5, because the average of all the numbers that come up converges to 3.5 as the number of rolls approaches infinity.

Let us perform a small simulation on rolling a die:

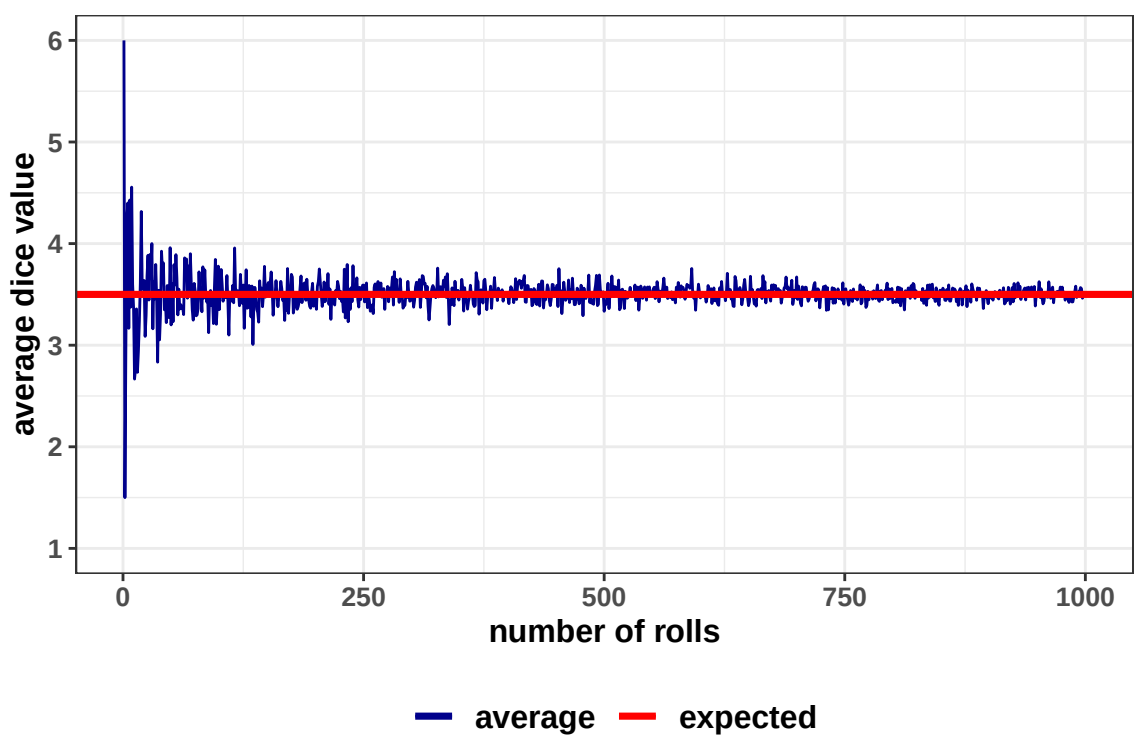


Figure 1: An example of the average die value behavior when number of rolls increases.

As we can see, the larger the number of trials, the closer the average value to the expected value.

For any two random variables X, Y, and for any two real numbers a, b,

$$\mathrm{E}\left[aX \pm bY\right] = a\mathrm{E}\left[X\right] \pm b\mathrm{E}\left[Y\right].$$

Variance

Variance (σ^2) is the expectation of the squared deviation of a random variable from its expected value:

$$\sigma^2 = \mathrm{V}[X] = \mathrm{E}\left[(X - \mathrm{E}[X])^2\right] = \mathrm{E}\left[X^2\right] - (\mathrm{E}[X])^2 .$$

Informally, it measures how far a set of (random) numbers are spread out from their average value.

The variance is also known as the ***second central moment*** (μ_2).

For discrete random variable X,

$$\sigma^2 = \mathrm{V}[X] = \sum_i (x_i - \mu)^2 p_i = \sum_i x_i^2 p_i - \mu^2.$$

For any two random variables X, Y, and for any two real numbers a, b,

$$\mathrm{V}[aX \pm bY] = a^2 \mathrm{V}[X] + b^2 \mathrm{V}[Y] .$$

Standard deviation

Standard deviation (σ) is a measure of the amount of variation of a random variable. A low standard deviation indicates that values of a random variable tend to be close to the expected value (μ), while a high standard deviation indicates that the values are spread out over a wider range.

The standard deviation is calculated as the square root of the variance:

$$\sigma = \mathrm{D}[X] = \sqrt{\mathrm{V}[X]} = \sqrt{\sum_i (x_i - \mu)^2 p_i} = \sqrt{\sum_i x_i^2 p_i - \mu^2}.$$

A useful property of the standard deviation is that, unlike the variance, it is expressed in the same units as the data.

Coefficient of variation

Coefficient of variation (CV), also known as relative standard deviation (RSD), is a standardized measure of dispersion of a random variable. It is often expressed as a percentage, and is defined as the ratio of the standard deviation σ to the expected value μ (or its absolute value, $|\mu|$):

$$CV = \frac{\sigma}{\mu}.$$

Skewness

Skewness is a measure of the asymmetry of random variable about its mean. The skewness value can be positive or negative, or undefined.

For a *unimodal* random variable, negative skew commonly indicates that the *tail* is on the left side of the distribution, and positive skew indicates that the *tail* is on the right.

The skewness of a random variable X is the third standardized moment γ_1, defined as

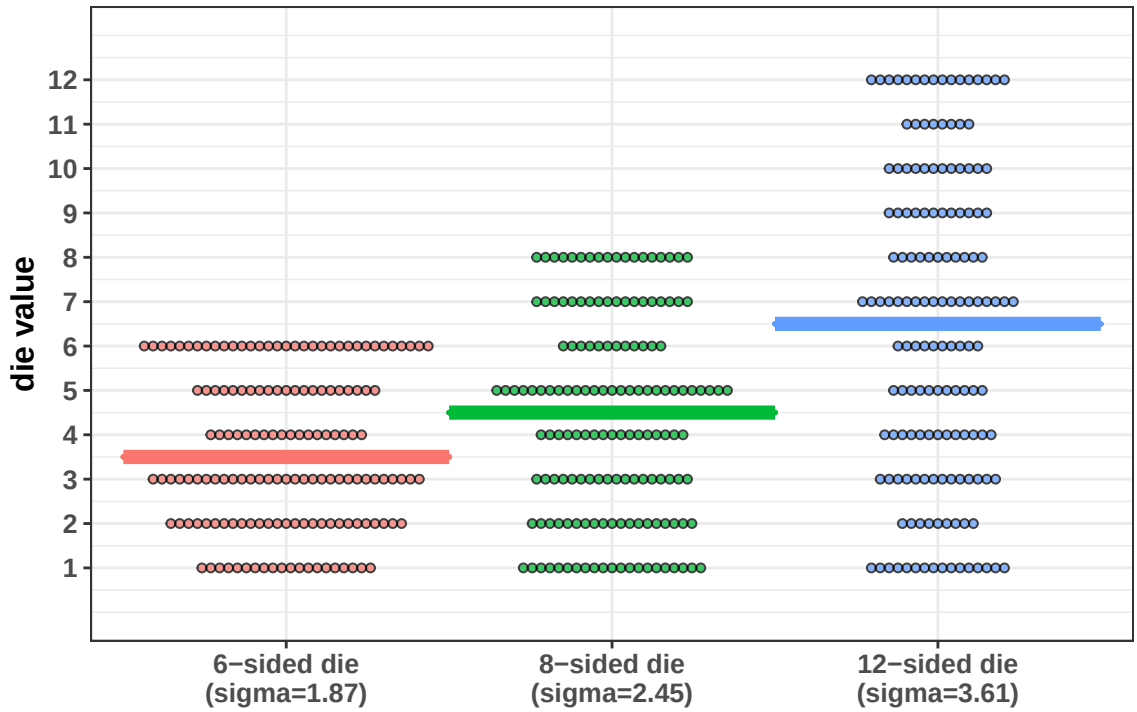


Figure 2: An example of the standard deviation impact on the data spread around the mean for 3 types of fair dice.

$$\gamma_1 = \mathrm{E}\left[\left(\frac{X-\mu}{\sigma}\right)^3\right] = \frac{\mathrm{E}\left[(X-\mu)^3\right]}{\sigma^3} = \frac{\mu_3}{\sigma^3},$$

where $\mu_3 = \mathrm{E}\left[(X-\mu)^3\right]$ is called a ***third central moment***.

Kurtosis

Kurtosis (from Greek: kyrtos or kurtos, meaning "curved, arching") is a measure of the "tailedness" of a random variable.

The kurtosis of a random variable X is the fourth standardized moment defined as

$$\mathrm{Kurt}\left[X\right] = \mathrm{E}\left[\left(\frac{X-\mu}{\sigma}\right)^4\right] = \frac{\mathrm{E}\left[(X-\mu)^4\right]}{\sigma^4} = \frac{\mu_4}{\sigma^4},$$

where $\mu_4 = \mathrm{E}\left[(X-\mu)^4\right]$ is called a ***fourth central moment***.

"Popular" discrete random variables

Bernoulli (binary) RV

- A random trial has two possible outcomes $\{\omega_1 = \text{"success"}, \omega_2 = \text{"fail"}\}$.

- We may consider a RV $X(\omega): \begin{cases} X(\omega_1 = \text{"success"}) = 1 \\ X(\omega_2 = \text{"fail"}) = 0 \end{cases}$

If probability of "success" is p, then probability of "fail" is $1-p$. Then a ***Bernoulli random variable*** X is defined as

$$X \sim \text{Ber}(p),\ X = \begin{cases} 1, & \text{with probability } p \\ 0, & \text{with probability } 1-p \end{cases}$$

- $\mathrm{E}[X] = \mu = 1 \cdot p + 0 \cdot (1-p) = p$
- $\mathrm{V}[X] = \sigma^2 = 1^2 \cdot p + 0^2 \cdot (1-p) - p^2 = p - p^2 = p(1-p)$
- $\mathrm{D}[X] = \sigma = \sqrt{\sigma^2} = \sqrt{p(1-p)}$

Discrete uniform RV

- A random trial has n possible outcomes $\{1, 2, \ldots, n\}$.
- Probabilities of each outcome are the same and equal to $\frac{1}{n}$.

Then, a ***Discrete uniform random variable*** X is defined as

$$X \sim DU(n),\ \mathrm{P}(X = k) = \frac{1}{n},\ k = 1, 2, \ldots, n.$$

- $\mathrm{E}[X] = \mu = \sum_{k=1}^{n} k \cdot \frac{1}{n} = \frac{1}{n}(1 + 2 + \ldots + n) = \frac{1}{n}\frac{(1+n)n}{2} = \frac{n+1}{2}$
- $\mathrm{V}[X] = \sigma^2 = \sum_{k=1}^{n} k^2 \cdot \frac{1}{n} = \frac{1}{n}\left(1^2 + 2^2 + \ldots + n^2\right) = \frac{1}{n}\frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6}$
- $\mathrm{D}[X] = \sigma = \sqrt{\sigma^2} = \sqrt{\frac{(n+1)(2n+1)}{6}}$

Counting random variables

Definition

Counting random variables *are the special types of random variables, and they describe the number of occurrences of a particular event.*

The commonly used types of counting random variables are:

- ***Binomial***.
- ***Negative binomial***.
- ***Poisson***.
- ***Hyper geometric***.
- ***Geometric***.

Binomial RV

- An experiment is performed n times.
- In each experiment, an event A ("success") may occur. It is an event of interest.
- The probability $\mathrm{P}(A)$ of observing A in one experiment equals to p.
- One is interested in a random variable $X =$ "number of success out of n" $(0 \leq X \leq n)$.

Then, $X \sim \text{Bin}\,(n, p)$, and

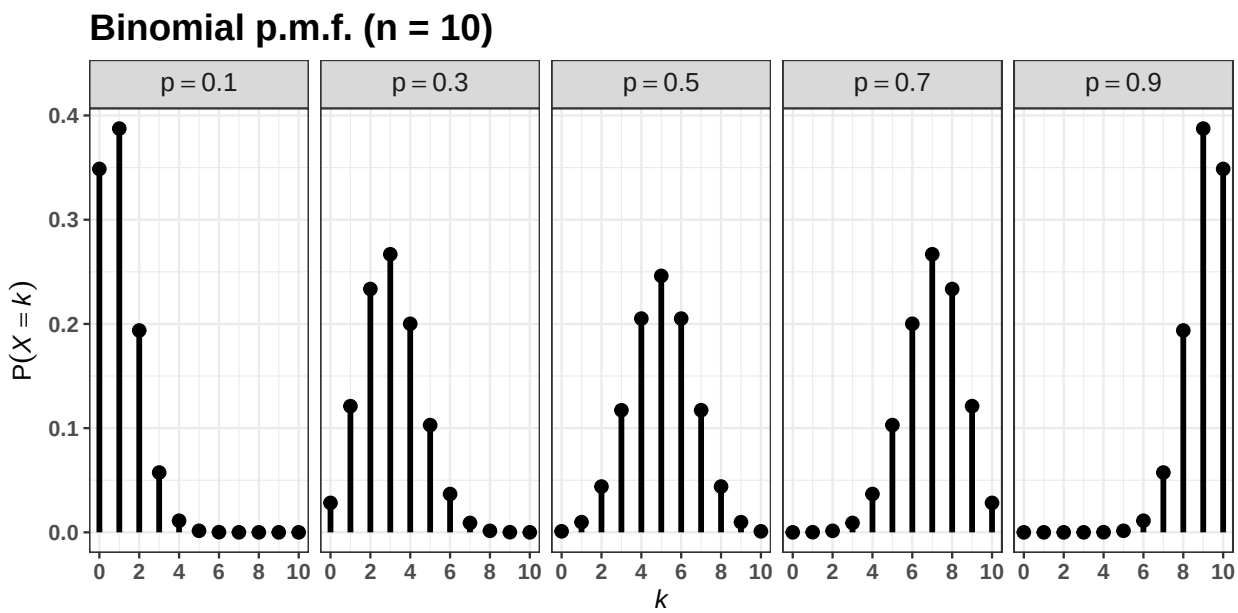
$$\text{P}\,(X = k) = \binom{n}{k} p^k (1-p)^{n-k},\ k = 0, 1, \ldots, n.$$

Here, $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is a *binomial coefficient*, and $n! = 1 \cdot 2 \cdot \ldots \cdot n$, $0! = 1$.

If $X_1, X_2, \ldots, X_n$ are *independent* Bernoulli (binomial) random variables, i.e. $X_i \sim Ber(p),\ i = 1, 2, \ldots, n$, then

$$X = X_1 + X_2 + \ldots + X_n \sim Bin(n, p).$$

- $\text{E}\,[X] = \mu = np$
- $\text{V}\,[X] = \sigma^2 = np(1-p)$
- $\text{D}\,[X] = \sigma = \sqrt{\sigma^2} = \sqrt{np(1-p)}$



Example (Tossing a coin).

One is tossing a *"fair"* coin 5 times. What is the probability of getting exactly 3 *"heads"*?

Solution:

- An experiment = "tossing a coin once".
- $n = 5$.
- $A = \{\text{"head"}\}$.
- $\text{P}\,(A) = p = \frac{1}{2}$.
- A random variable X = "number of "heads" out of 5".
- The answer is

$$\text{P}\,(X = 3) = \binom{5}{3} \left(\frac{1}{2}\right)^3 \left(1 - \frac{1}{2}\right)^{5-3} = \frac{5!}{3!2!} \left(\frac{1}{2}\right)^5 \approx 0.3125.$$

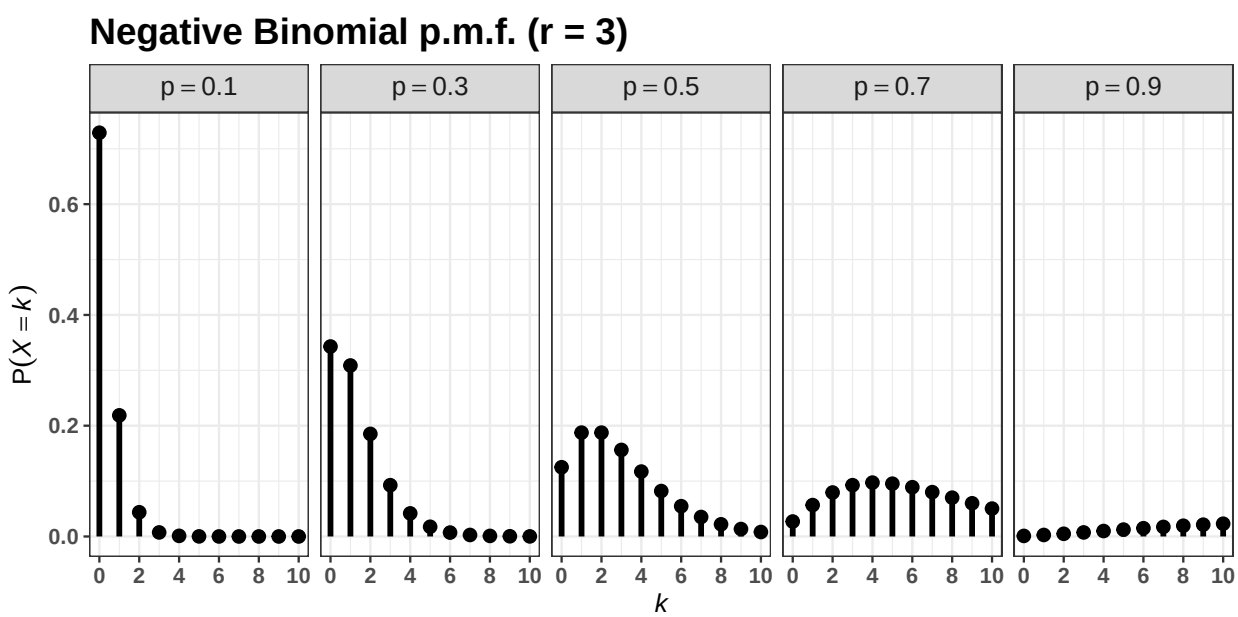
Negative binomial RV

- An experiment is performed *some number of* times (it depends on the trial setup).
- In each experiment, either an event A ("success") or an event B ("failure") may occur. A is an event of interest.
- The probability $\mathrm{P}(A)$ of observing A in one experiment equals to p, and the probability $\mathrm{P}(B)$ of observing B equals to $1-p$.
- One is interested in a random variable $X =$ "number of success until r failures occur" ($X = 0, 1, 2, \ldots$).

Then, $X \sim \mathrm{NegBin}(r, p)$, and

$$\mathrm{P}(X = k) = \binom{k+r-1}{k} p^k (1-p)^r, \ k = 0, 1, 2, \ldots.$$

- $\mathrm{E}[X] = \mu = \frac{pr}{1-p}$
- $\mathrm{V}[X] = \sigma^2 = \frac{pr}{(1-p)^2}$
- $\mathrm{D}[X] = \sigma = \sqrt{\sigma^2} = \frac{\sqrt{pr}}{1-p}$



Example (tossing a coin)

One is tossing a *"fair"* coin 5 times. What is the probability of getting more than 2 *"heads"* before 3 *"tails"* occur?

Solution:

- An experiment = "tossing a coin once".
- $r = 3 =$ number of "tails" ("failures" in this case) .
- $A = \{$"head"$\}$.
- $\mathrm{P}(A) = p = \frac{1}{2}$.
- A random variable $X =$ "number of "heads" before 3 "tails" ".
- The answer is

$$
\begin{aligned}
\mathrm{P}(X>2) &= 1-\mathrm{P}(X\le 2) = 1-\mathrm{P}(X=0)-\mathrm{P}(X=1)-\mathrm{P}(X=2) = \\
&= 1-\binom{0+3-1}{0}\left(\frac{1}{2}\right)^0\left(1-\frac{1}{2}\right)^3 - \\
&- \binom{1+3-1}{1}\left(\frac{1}{2}\right)^1\left(1-\frac{1}{2}\right)^3 - \\
&- \binom{2+3-1}{2}\left(\frac{1}{2}\right)^2\left(1-\frac{1}{2}\right)^3 = \\
&= 1-\binom{2}{0}\left(\frac{1}{2}\right)^3 - \binom{3}{1}\left(\frac{1}{2}\right)^4 - \binom{4}{2}\left(\frac{1}{2}\right)^5 = \\
&= 1-\frac{2!}{0!2!}\frac{1}{8} - \frac{3!}{1!2!}\frac{1}{16} - \frac{4!}{2!2!}\frac{1}{32} \approx 0.5.
\end{aligned}
$$

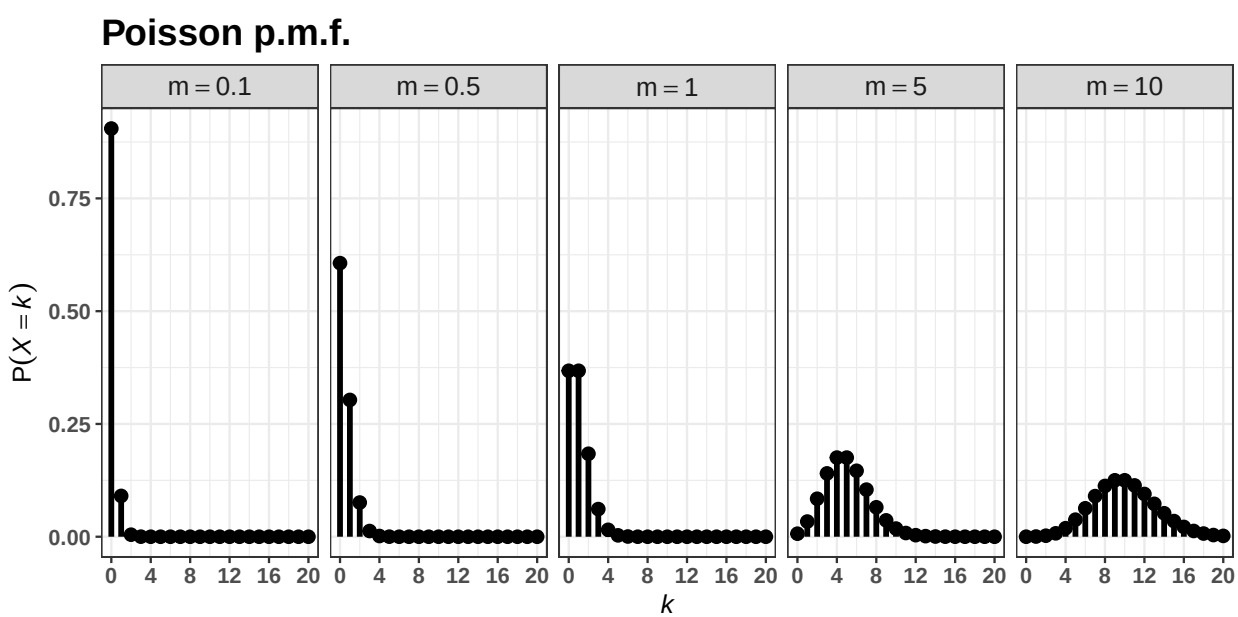
Poisson RV

- An experiment is performed n times (where n is ***sufficiently large***, $n \to +\infty$).
- In each experiment, an event A ("success") may occur. It is an event of interest.
- The probablity $\mathrm{P}(A)$ of observing A in one experiment equals to p, and it is ***sufficiently small***, i.e. the ***event is rare***.
- One is interested in a random variable $X =$ "number of success".

Then, $X \sim \mathrm{Po}(m)$, and

$$
\mathrm{P}(X=k) = \frac{m^k}{k!}e^{-m},\ m = np;\ k = 0, 1, \ldots.
$$

- $\mathrm{E}[X] = \mu = m$
- $\mathrm{V}[X] = \sigma^2 = m$
- $\mathrm{D}[X] = \sigma = \sqrt{\sigma^2} = \sqrt{m}$



Example (number of left-handed people in a population)

The probability of being left-handed equals to 0.1. What is the probability that there are at least 2 left-handed people in a group of 50?

Solution:

- An experiment = "select a person".
- $n = 50$ (can be considered as ***large***).
- $A = \{$"a person is left-handed"$\}$.
- $\mathrm{P}(A) = p = 0.1$ (can be considered as ***small***).
- A random variable X = "number of left-handed people out of 50".
- $m = np = 5$.

$$\begin{aligned}\mathrm{P}(X \geq 2) &= 1 - \mathrm{P}(X = 0) - \mathrm{P}(X = 1) \\ &= 1 - \frac{5^0}{0!}e^{-5} - \frac{5^1}{1!}e^{-5} \approx 0.9595723.\end{aligned}$$

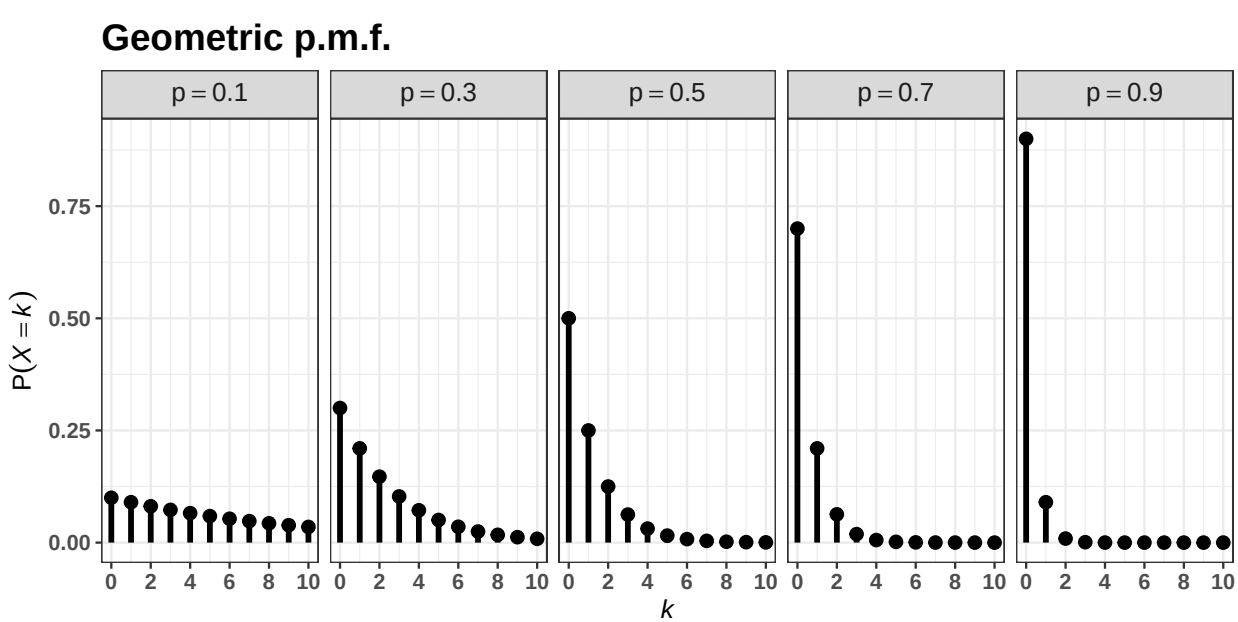
Geometric RV

- In each experiment, an event A ("success") may occur. It is an event of interest.
- The probability $\mathrm{P}(A)$ of observing A in one experiment equals to p.
- An experiment is performed until first "success" is observed.
- One is interested in a random variable X = "number of "fails" before a "success" is observed".

Then, $X \sim \mathrm{Ge}(p)$, and

$$\mathrm{P}(X = k) = p(1-p)^k, \ k = 0, 1, \ldots.$$

- $\mathrm{E}[X] = \mu = \frac{1}{p}$
- $\mathrm{V}[X] = \sigma^2 = \frac{1-p}{p^2}$
- $\mathrm{D}[X] = \sigma = \sqrt{\sigma^2} = \frac{\sqrt{1-p}}{p}$



Example.

Products produced by a machine have a 3% defective rate. What is the probability that the first defect occurs in the fifth item inspected?

Solution:

- $A = \{$"an item has a defect"$\}$.
- $\mathrm{P}(A) = p = 0.03$.
- A random variable $X =$ "number of "good" items investigated before the defective one".
- $k = 4$
- The answer is

$$\mathrm{P}(X = 4) = 0.03 \cdot (1 - 0.03)^4 \approx 0.0265588.$$

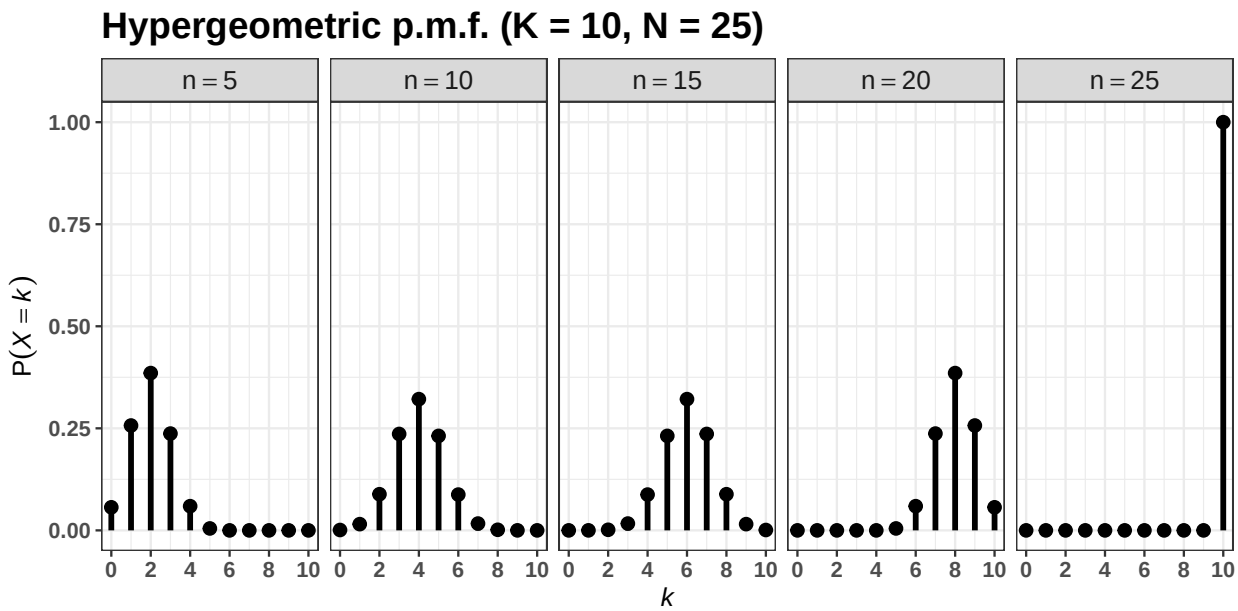
Hypergeometric RV

- In each experiment, either an event A ("success") or an event event $B = A^c$ ("fail") may occur.
- The probability $\mathrm{P}(A)$ of observing A changes after each time the experiment is performed.
- An experiment is performed n times ($n \leq N$, N is the population size).
- K is the number of all possible "successes" out of N ($K \leq N$).
- One is interested in a random variable $X =$ "number of "successes" observed out of" n experiments.

Then, $X \sim \mathrm{HGe}(N, K, n)$, and

$$\begin{aligned} \mathrm{P}(X = k) &= \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}, \\ N &= 0, 1, \ldots, \\ K &= 0, 1, \ldots, N, \\ n &= 0, 1, \ldots, N, \\ k &= 0, 1, \ldots, K. \end{aligned}$$

- $\mathrm{E}[X] = \mu = n\frac{K}{N}$
- $\mathrm{V}[X] = \sigma^2 = n\frac{K}{N}\frac{N-K}{N}\frac{N-n}{N-1}$
- $\mathrm{D}[X] = \sigma = \sqrt{\sigma^2} = \sqrt{n\frac{K}{N}\frac{N-K}{N}\frac{N-n}{N-1}}$



Example.

In a population of 20 subjects, 15 subjects are younger than 25 years. 7 subjects are selected at random to participate in a (whatever) test. What is the probability that there are exactly 5 subjects younger than 25 years?

Solution:

- $A = \{$"a subject is younger than 25 years"$\}$.
- A random variable $X =$ "number of "young" subjects in a group of 7 selected from the population of size 20".
- $N = 20$
- $K = 15$
- $n = 7$
- $k = 5$
- The answer is

$$\mathrm{P}\left(X = 5\right) = \frac{\binom{15}{5}\binom{20-15}{7-5}}{\binom{20}{7}} \approx 0.3873839.$$

Connections between distributions

Law of Small Numbers

Reference: L. Bortkiewicz, *Das Gesetz der kleinen Zahlen*, Leipzig: B.G. Teubner (1898).

If an experiment is carried out by n independent trials, and the probability for "success" in each trial equals to p, then the number of successes X is given by the binomial distribution:

$$X \sim \mathrm{Bin}\,(n, p)\,.$$

If $n \to +\infty$ and $p \to 0$ so that $m = n \cdot p$ is constant, we have approximately that

$$X \sim \mathrm{Po}\,(m)\,.$$

Negative binomial and Geometric RV's

$$\mathrm{Ge}(p) = \mathrm{NegBin}(1, 1-p)$$

Hypergeometric and Bernoulli RV's

$$\mathrm{HGe}(N, K, 1) = \mathrm{Ber}\left(\frac{K}{N}\right)$$